

Contestant Number: _____

Time: _____

Rank: _____



FUNDAMENTAL SPREADSHEET APPLICATIONS (230)

STATE 2021

Production:

Job 1: Create Spreadsheet with Formulas _____ (175 points)

Job 2: Format Spreadsheet _____ (135 points)

Job 3: Create Pivot Table and Pivot Table Chart _____ (50 points)

TOTAL POINTS _____ (360 points)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

GENERAL INSTRUCTIONS:

1. Put your contestant number and printout number in the right section of the footer on each printout unless instructed otherwise. Your name or initials should *not* appear on any work you submit.
2. If you finish before the end of the testing time, notify the proctor. Time may be a factor in determining the winner in the event of a tie.
3. When turning in your contest, the jobs should be arranged in printout order.

SCENARIO:

You are a student aide in the front office of BPA HS. Your job today is to create a spreadsheet for one of the teachers that contains data on hall pass usage for her classes for the month of November.

GRADER: Points are all or none unless otherwise noted!

Job 1 (Worksheet with Formulas)	Points Possible	Points Earned
Contestant # and Job # in right section of footer	10	
11 pt. Times New Roman font used for data	10	
Set to Landscape Orientation	5	
Spreadsheet information entered	10	
Formula entered in cell G3 that calculates the total time each student is out of class (Answers will not be in time format – that formatting takes place in Job 2).	10	
Formula copied down to row 20	10	
Updated spreadsheet with the December information (rows 21-24)	10	
Updated the formulas to include the December data	10	
The following information was entered in the cells listed: • A26: How many passes to: • A27: Bathroom • A28: Office • A29: Nurse	20	
Function entered in Cell B27 that counts the number of bathroom visits	10	
Function entered in Cell B28 that counts the number of office visits	10	
Function entered in Cell B29 that counts the number of nurse visits	10	
In cell D27 entered the following information: “longest time out of class for one student”	5	
In cell G27, entered a function that finds the longest time a student was out of class	10	
Renamed the worksheet Mr. Norris	5	
Centered vertically and horizontally on one page	10	
Printout is in landscape view with the worksheet centered on the page	10	
Printout of formulas in landscape view with the worksheet centered on one page	10	
Subtotal	175	

Job 2 (Formatted Worksheets)	Points Possible	Points Earned
Contestant # and Job # in right section of footer	10	
Printed in landscape, centered vertically and horizontally on one sheet	10	
Formatted A1 to merge center across the data	10	
Increased the font size of the title to 20	5	
Placed a thick bottom border under the title	5	
Formatted G3:B27 to "time" with XX-XX-XX format	10	
Bolded the column titles (Date, In, Location, Etc.)	10	
G3:G24 use conditional formatting as follows: - Icon Set 3 arrows (gray) - Up arrow: >0:15:00 - Middle arrow: 0:10:00 <i>(After typing in the information, it will convert the time to a number)</i>	15	
Created a second conditional formatting on the same cell locations to highlight in yellow for any times greater than 0:15:00 minutes.	10	
Made a copy of your worksheet and renamed the new worksheet: Mr. Norris subtotal	5	
On the new worksheet, do a custom sort on first name	5	
Subtotal the data, change in first name, function count and subtotal on the total only	15	
Updated the function in cell G34 to include the total time cells for students, not the count cells	10	
Formatted this page to only print on one sheet at 93% normal size	10	
Printout of the two formatted spreadsheets centered horizontally and vertically on one page	5	
Subtotal	135	

Job 3 (Pivot Table and Pivot Chart)	Points Possible	Points Earned
Contestant # and Job # in right section of footer	10	
Based on Mr. Norris spreadsheet, a Pivot Table and Pivot Chart were created with the following fields: • Location: Legend series • First Name: Axis category • Total: Values (minutes and hours off)	15	
Placed in new worksheet; renamed Pivot Table and Chart	10	
Chart moved so the top left corner is in Cell B13	5	
Centered vertically and horizontally on one page	10	
Subtotal	50	
TOTAL POINTS	360	

JOB 1: Create Spreadsheet with Formulas

1. Enter the following data into an Excel spreadsheet like shown below, and complete the calculations as instructed. Use Times New Roman size 11 font and landscape orientation.

BPA HS - Hall Pass Summary						
Date	In Location	First Name	Last Name	Out Time	In Time	Total
2019-11-06	Bathroom	Asia	Williams	11:00:13	11:07:50	
2019-11-06	Bathroom	Ned	Jones	09:26:34	09:37:59	
2019-11-07	Bathroom	Kammi	Youngstown	09:32:20	09:44:21	
2019-11-08	Nurse	Kammi	Youngstown	09:15:50	09:22:46	
2019-11-11	Bathroom	John	Clark	09:20:41	09:27:31	
2019-11-12	Nurse	John	Clark	09:21:02	09:24:08	
2019-11-12	Nurse	Ned	Jones	09:36:48	09:40:37	
2019-11-13	Bathroom	Grace	Smith	09:33:22	09:40:50	
2019-11-13	Bathroom	Kammi	Youngstown	09:09:49	09:17:52	
2019-11-14	Nurse	Asia	Williams	10:40:07	10:52:27	
2019-11-14	Bathroom	John	Clark	09:22:12	09:28:03	
2019-11-15	Bathroom	John	Clark	09:01:16	09:10:15	
2019-11-18	Office	Asia	Williams	11:12:10	11:22:12	
2019-11-19	Bathroom	John	Clark	09:26:07	09:32:55	
2019-11-20	Office	Kammi	Youngstown	09:22:55	09:33:04	
2019-11-21	Bathroom	Brandon	MacKenzie	01:14:54	01:20:06	
2019-11-21	Bathroom	Ned	Jones	09:10:34	09:16:19	
2019-11-22	Office	Kammi	Youngstown	08:39:31	09:03:11	

2. Create a formula in cell G3 that calculates the total time each student was out of class. (*The answers will not be in a “time” format that will be done in Job 2).*
3. Copy this formula down to row 20.
4. You have to update the spreadsheet with the first week of December data also. Please enter the new information at the end of the spreadsheet. (rows 21-24)

2019-12-02	Bathroom	Ned	Jones	09:36:52	09:44:50
2019-12-03	Bathroom	Brandon	MacKenzie	01:32:08	01:39:15
2019-12-03	Office	John	Clark	09:20:00	09:23:30
2019-12-03	Nurse	Ned	Jones	09:01:52	09:19:58

5. Update the formula in the spreadsheet for the new information you just entered.
6. In cell A26 enter the following information: “How many passes to:”
7. In cell A27 enter the following information: “Bathroom”.
8. In cell A28 enter the following information: “Office”.
9. In cell A29 enter the following information: “Nurse”

10. In cell B27 enter a function that counts the number of Bathroom visits for all students.
11. In cell B28 enter a function that counts the number of Office visits for all students.
12. In cell B29 enter a function that counts the number of Nurse visits for all students.
13. In cell D27, enter the following information: “Longest time out:”
14. In cell G27, enter a function that finds the longest time a student was out of class.
15. Rename the worksheet tab to Mr. Norris.
16. Print a copy of the worksheet vertically and horizontally centered, in landscape view, with gridlines, and adjust the column widths to show all data on one page.
17. Print a copy of the worksheet showing the formulas vertically and horizontally centered, in landscape view, with gridlines, and scale to one page.

JOB 2: Format the Spreadsheet you created in Job 1 using the following instructions:

1. Merge and center the data in row 1 from A:G. Increase the font size to 20.
2. Place a thick bottom border under the title.
3. Format G3:G24 and G27 to “time” with XX:XX:XX format.
4. Indent cells A27:A29 over two times.
5. Bold the column titles (Date, in, location, etc.).
6. G3:G24 use conditional formatting with the following set rules:
 - Icon Set 3 arrows (gray)
 - Up arrow: $\geq 0:15:00$ *(After typing in the information, it will convert the time to a number)*
 - Middle arrow: 0:10:00
7. Create a second conditional formatting on the same cells to apply a yellow fill to any times greater than 0:15:00 minutes.
8. Make a copy of your worksheet. Rename the new worksheet: Mr. Norris _ subtotal.
9. On the new worksheet, do a custom sort on first name.
10. Subtotal the data, change in first name, function count and subtotal on the total only.
Everything else should remain the same.
11. Update the function in cell G34 to exclude the count.
12. Format this page to only print on one sheet at 93% normal size.
13. Print out the two formatted spreadsheets with them centered horizontally and vertically on one page.

JOB 3: Create a Pivot Table and Pivot Chart

Create a Pivot Chart and Pivot Table from the Mr. Norris spreadsheet you have created in the previous jobs showing the breakdown for each student at each location. It should be moved to its own sheet. Change the sheet name to Pivot Table and Chart.

Format the chart to look like the following:

1. Back on the original spreadsheet (Mr. Norris); create a PivotTable and Pivot Chart in a new worksheet.
2. Add three fields; in **Location** in legend (series); **First Name** in Axis (category), and **Total** in Values (*turn off minutes and hours*).
3. Rename the chart tab **Pivot Table and Chart**.
4. Move the chart so that the top left corner is in Cell B13.
5. Print on one page, vertically and horizontally centered.



You should have the following five printouts:

Job 1--unformatted

Job 1--formulas

Job 2--formatted with subtotals

Job 2--formatted

Job 3--Pivot Table and Pivot Chart